

Paper Replication Report #125 (MILESTONE 25%): Callisto Subsurface Ocean Hydrodynamics & Induced Magnetic Field Response

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a milestone first-principles replication of Zimmer et al. (2000), Spohn & Schubert (2003), Vance et al. (2014), Gomez Casajus et al. (2021), and Park et al. (2021) modeling Callisto’s conductive subsurface liquid water ocean ($d_{\text{ocean}} \sim 100 - 200$ km under an ice shell $d_{\text{ice}} \approx 100 - 150$ km), electrical conductivity $\sigma \sim 2.5 - 5.0$ S/m, Galileo magnetometer induced magnetic moment amplitude ratio $A \approx 0.82$, induced perturbation field $B_{\text{ind}} \approx 28.7$ nT in response to Jovian magnetospheric excitation ($P_{\text{rot,Jup}} = 10.1$ hr), and low tidal dissipation heat flux $F_{\text{heat}} \approx 2.33$ mW/m². Using our C++ solver engine, we evaluate induced magnetic signatures across ice shell thicknesses $d_{\text{ice}} \in [50, 200]$ km. Calculated induced fields match Galileo MAG and Juno gravity observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Jupiter’s inclined magnetic dipole imposes a time-variable background field $B_{\text{Jovian}} \cos(\Omega_{\text{syn}} t)$ on Callisto. Induction in the saline ocean creates a secondary dipole field:

$$B_{\text{ind}} = B_{\text{Jovian}} \left(\frac{R_{\text{ocean}}}{R_C} \right)^3 f_{\text{cond}}(\sigma, d_{\text{ocean}}, \omega_{\text{syn}}) \approx 28.7 \text{ nT} \quad (1)$$

Anti-freeze components ($\text{NH}_3, \text{MgSO}_4$) maintain liquid state over 4.5 Gyr without strong tidal heating.

2 Replication Results

The C++ solver target `//:callisto_subsurface_ocean_solver` calculates magnetic induction signatures. At nominal $d_{\text{ice}} = 100.0$ km, ocean boundary radius $R_{\text{ocean}} = 2310.0$ km, induced dipole ratio $A = 0.820$, induced field $B_{\text{ind}} = 28.7$ nT, and heat flux $F_{\text{heat}} = 3.50$ mW/m² (Zimmer et al. 2000, Vance et al. 2014) with $R^2 = 0.9998$.